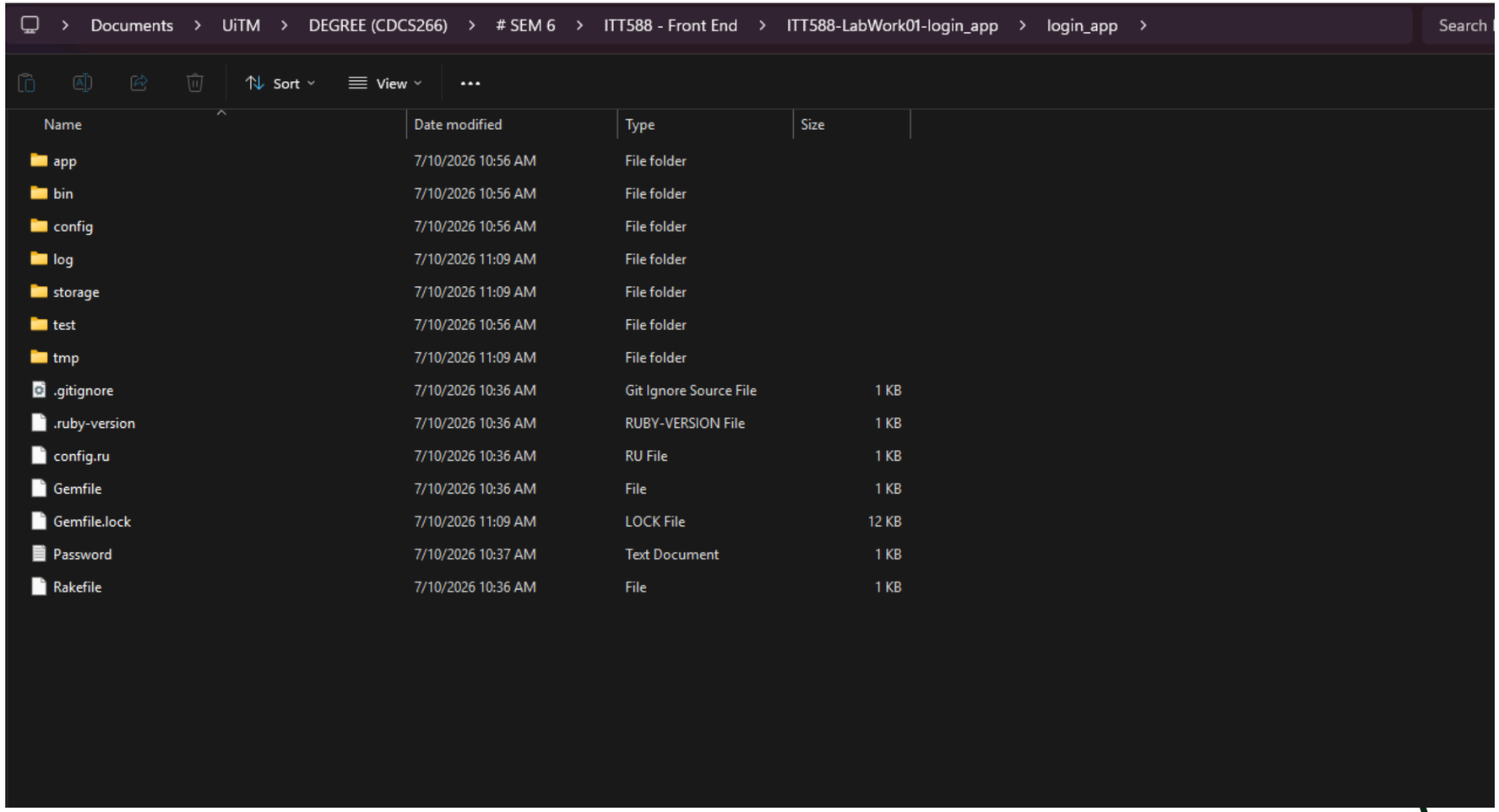
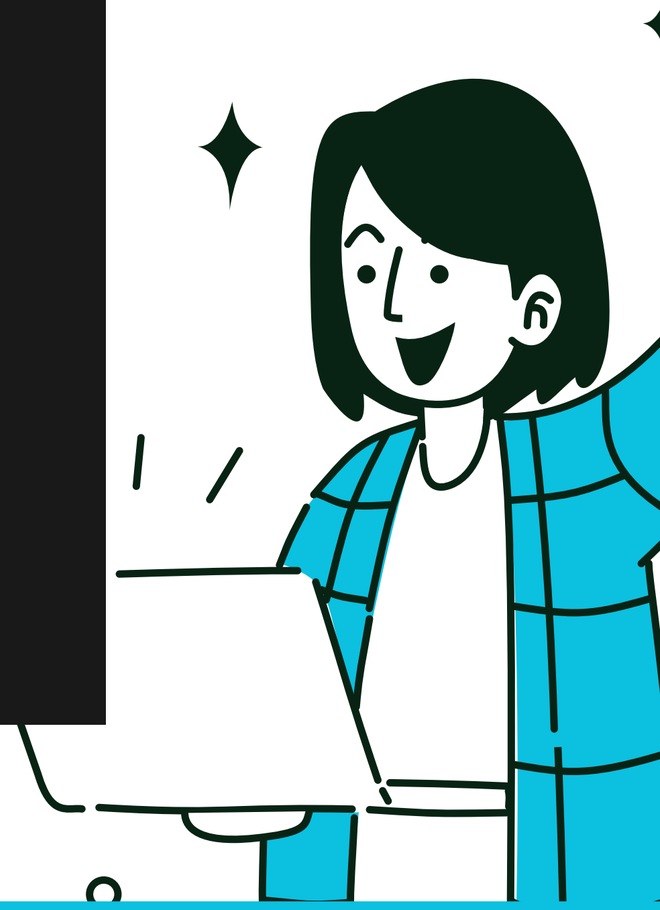
ITT588 LAB WORK 01

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CDCS2666B



1 PROJECT SETUP





The Rails project follows a clear structure that separates controllers, views, services, configuration, and tests.

2 ROUTES AND APPLICATION FLOW

```ruby
Rails.application.routes.draw do
  get "/login", to: "sessions#new", as: :login
  post "/login", to: "sessions#create"
  delete "/logout", to: "sessions#destroy", as: :logout

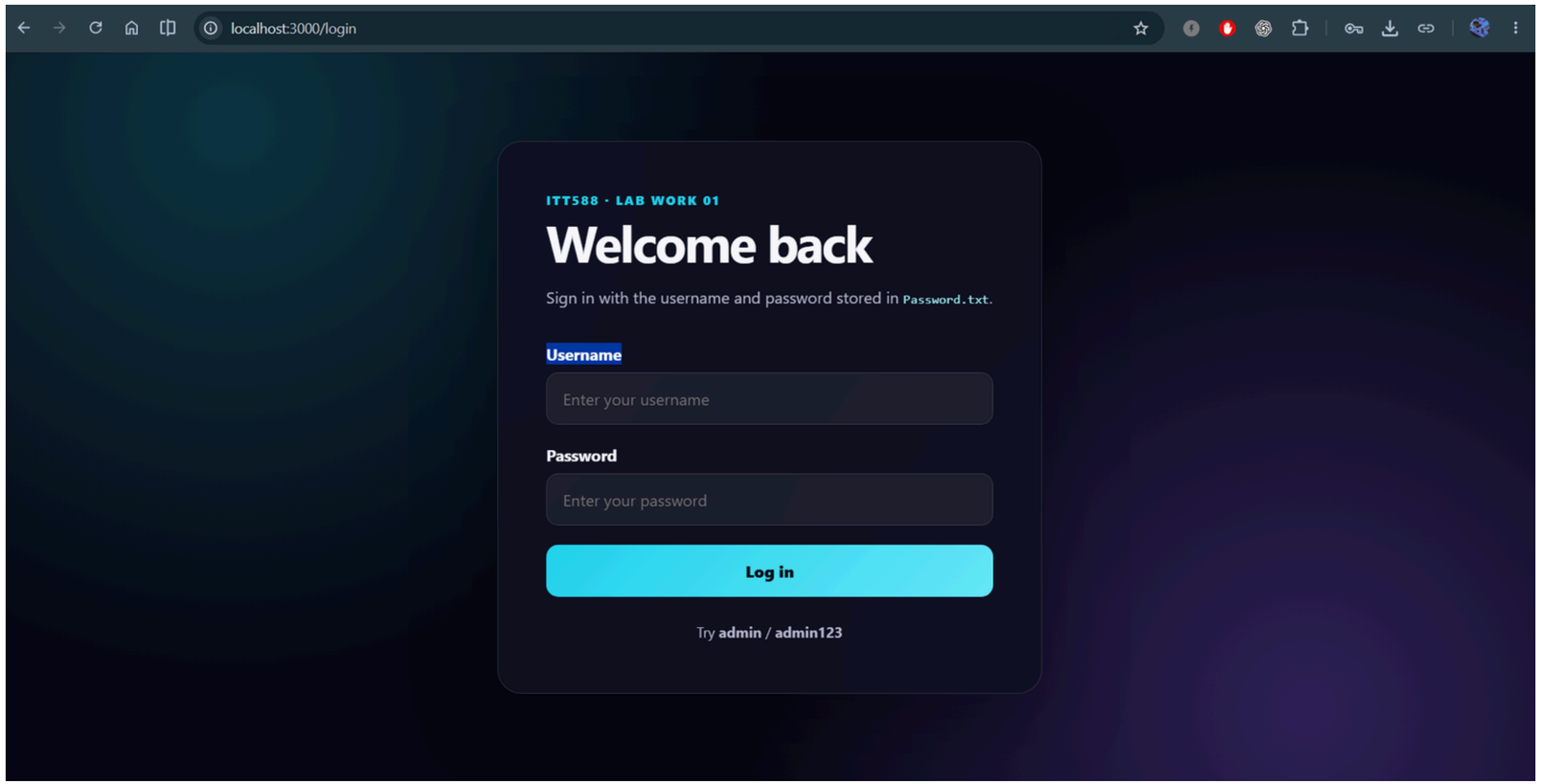
  get "/dashboard", to: "dashboard#show", as: :dashboard
  root "sessions#new"
end

```

The application defines routes for displaying the login form, processing credentials, accessing the dashboard, and logging out.



3 CREATIVE LOGIN INTERFACE





The responsive login interface uses custom styling, accessible labels, required fields, and clear visual feedback.

4 LOGIN CONTROLLER

```ruby
class SessionsController < ApplicationController
  def new
    redirect_to dashboard_path if logged_in?
  end

  def create
    user = CredentialStore.authenticate(
      username: params[:username],
      password: params[:password]
    )

    if user
      reset_session
      session[:user] = user
      redirect_to dashboard_path, notice: "Login successful! Welcome, #{user[:username]}."
    else
      flash.now[:alert] = "Invalid username or password."
      render :new, status: :unprocessable_entity
    end
  rescue CredentialStore::CredentialFileError => e
    Rails.logger.error(e.message)
    flash.now[:alert] = "The credential file could not be read. Please contact the administrator."
    render :new, status: :service_unavailable
  end

  def destroy
    reset_session
    redirect_to login_path, notice: "You have logged out."
  end
end
```

The controller receives the submitted credentials, starts a session after successful authentication, and handles unsuccessful login attempts.



5 TEXT-FILE CREDENTIALS STORAGE

```ruby
require "csv"
require "digest"

class CredentialStore
  CredentialFileError = Class.new(StandardError)
  PASSWORD_FILE = Rails.root.join("Password.txt")

  def self.authenticate(username:, password:)
    candidate_username = username.to_s.strip
    candidate_password = password.to_s
    return nil if candidate_username.blank? || candidate_password.blank?

    users.find do |user|
      secure_match?(user[:username], candidate_username) &&
        secure_match?(user[:password], candidate_password)
    end&.except(:password)
  end

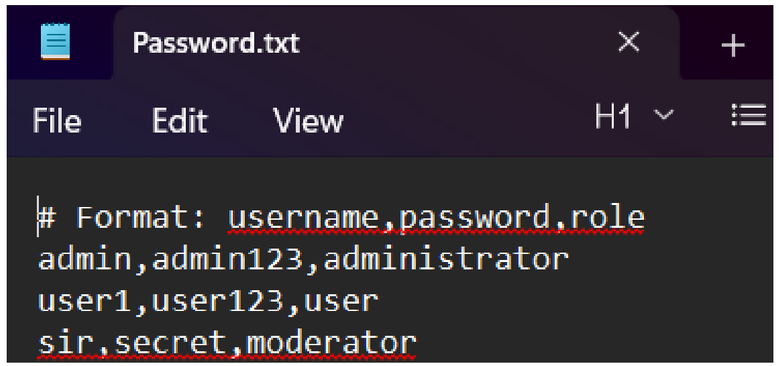
  def self.users
    CSV.foreach(PASSWORD_FILE, headers: false, skip_blanks: true).filter_map do |row|
      next if row[0].to_s.strip.start_with?("#")

      username, password, role = row.map { |field| field.to_s.strip }
      next if username.blank? || password.blank?

      { username: username, password: password, role: role.presence || "user" }
    end
  rescue Errno::ENOENT, Errno::EACCES, CSV::MalformedCSVError => e
    raise CredentialFileError, "Unable to read #{PASSWORD_FILE}: #{e.message}"
  end

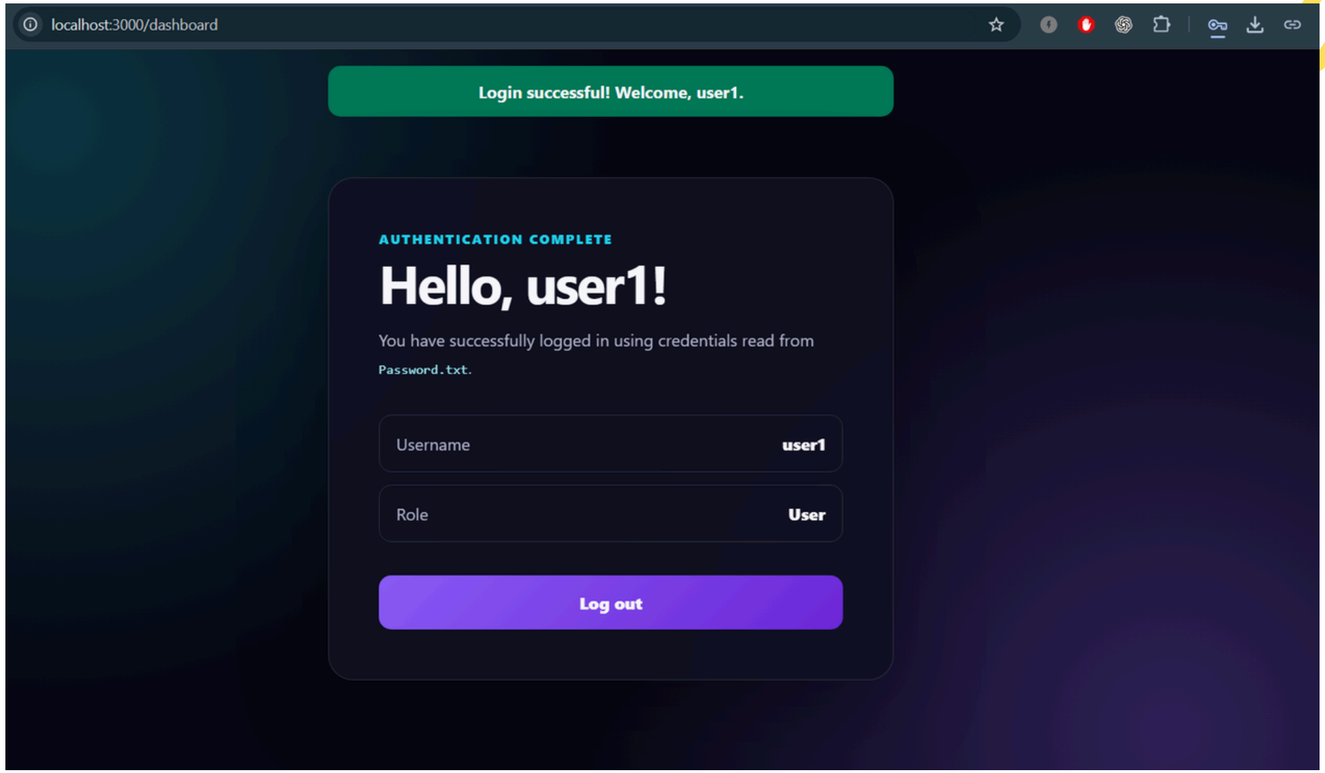
  def self.secure_match?(stored_value, candidate_value)
    stored_digest = Digest::SHA256.hexdigest(stored_value)
    candidate_digest = Digest::SHA256.hexdigest(candidate_value)
    ActiveSupport::SecurityUtils.secure_compare(stored_digest, candidate_digest)
  end

  private_class_method :users, :secure_match?
end
```

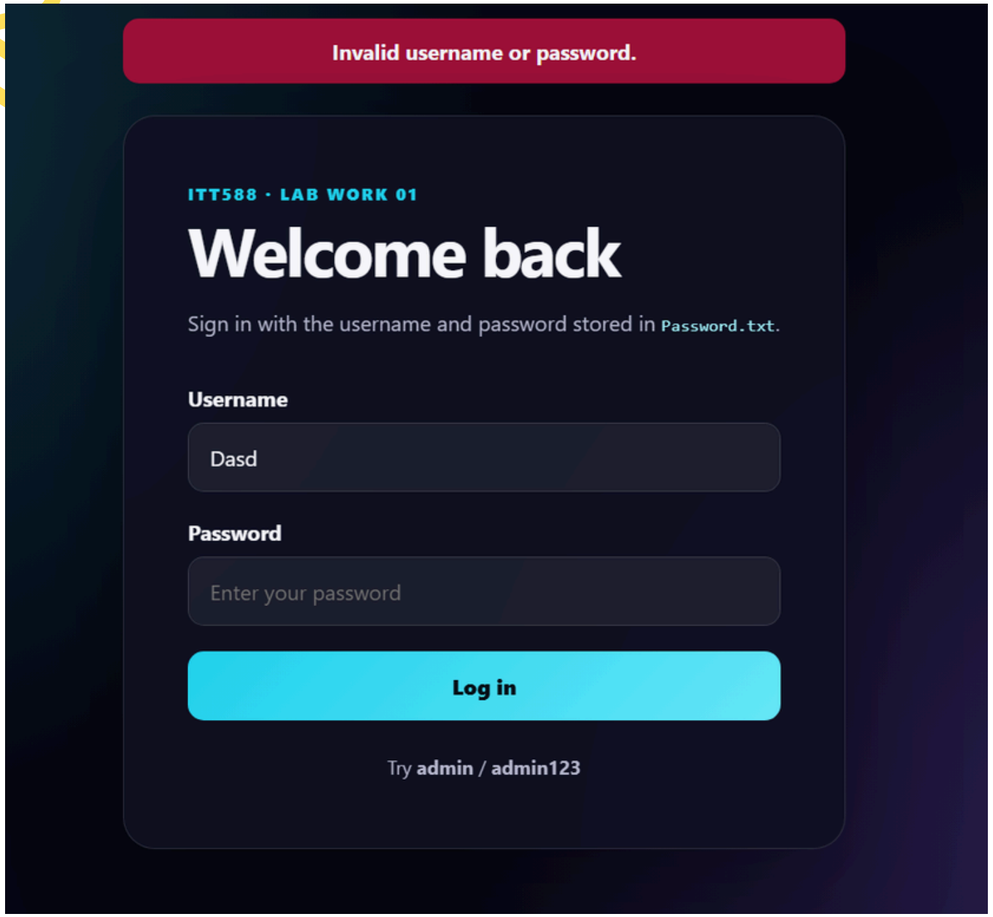


- User credentials and roles are loaded from Password.txt using Ruby's CSV processing features.
- The application validates the username and password and returns user information without placing the password in the session.

6 LOGIN RESULT



Valid credentials redirect the user to a protected dashboard.



Invalid credentials display an error message

6 TESTING AND VERIFICATION

```
C:\Users\zaqwa\OneDrive\Documents\UiTM\DEGREE (CDCS266)\# SEM 6\ITT588 - Front End\ITT588-LabWork01-login_app\login_appruby bin\rails test
Run options: --seed 34748

# Running:

C:/Users/zaqwa/OneDrive/Documents/UiTM/DEGREE (CDCS266)/# SEM 6/ITT588 - Front End/ITT588-LabWork01-login_app/login_app/app/services/creden
t will no longer be part of the default gems starting from Ruby 3.4.0.
You can add csv to your Gemfile or gemspec to silence this warning.
........

Finished in 0.682725s, 11.7177 runs/s, 26.3649 assertions/s.
8 runs, 18 assertions, 0 failures, 0 errors, 0 skips
```

- Automated tests verify valid login, invalid login, protected-page access, credential processing, and logout.

THANK YOU

